Higher order non-homogeneous LDE with constant coefficients:

Step ①

→ Given eqn → ①

→ L.H.S = 0

Let, $y = e^{mx}$

$y' = me^{mx}$

$y'' = m^2e^{mx}$

$\vdots$

→ put value in eqn ②

$(\quad) e^{mx} = 0$

$e^{mx} \neq 0$

m = values

∴ Distinct : $m = \alpha, \beta, \gamma$

$eq^n \Rightarrow y_1 = c_1 e^{\alpha x} + c_2 e^{\beta x} + c_3 e^{\gamma x}$

∴ Equal : $m = \alpha, \alpha, \alpha$

$eq^n \Rightarrow y_1 = c_1 e^{\alpha x} + c_2 x e^{\alpha x} + c_3 x^2 e^{\alpha x}$

∴ Complex : $m = \alpha \pm i\beta$

$eq^n \Rightarrow y_1 = e^{\alpha x}(c_1 \sin\beta x + c_2 \cos\beta x)$

$\boxed{y_1 = \_\_\_\_}$

Step ②

R.H.S

→ UC : $x^n \rightarrow \{x^n, x^{n-1}, x^{n-2}\}$

: $\sin ax \rightarrow \{\sin ax, \cos ax\}$

: $\cos ax \rightarrow \{\cos ax, \sin ax\}$

: $e^{ax} \rightarrow \{e^{ax}\}$

→ do Union to find S

→ put values in eqn ①

$y_2 = A \quad B \quad C$

$y_2' =$

$y_2'' =$

→ simplify

Compare L.H.S with R.H.S and take coefficients

→ Find A, B, C

→ $\boxed{y_2 = \_\_\_\_}$

Step-3

General sol^n is $\boxed{y = y_1 + y_2}$

$\boxed{y = \_\_\_\_}$

IF y & y' given find c_1 & c_2

(Ans)

1) Given,

$$y'' - 2y' - 3y = 2e^{x} - 10\sin x \quad \rightarrow (1)$$

Let, $y'' - 2y' - 3y = 0 \quad \rightarrow (2)$

Also. Let, $y = e^{mx}$ be a trial sol^n of (2)

$\therefore y' = me^{mx},\ y'' = m^2e^{mx}$

put these value in (2)

$\therefore (m^2 - 2m - 3)e^{mx} = 0$

since $e^{mx} \neq 0$, auxillary eq^n will be,

$$m^2 - 2m - 3 = 0$$

$\Rightarrow m^2 - 3m + m - 3 = 0$

$\Rightarrow (m+1)(m-3) = 0$

$\therefore m = -1, 3$ [distinct]

$\therefore y_1 = c_1e^{-x} + c_2e^{3x}$

Again from (1),

$S_1 = \{e^x\},\ S_2 = \{\sin x, \cos x\}$ $\boxed{\{\text{Union set}\}}$

$\therefore$ UC set,

$$S = \{e^x, \sin x, \cos x\}$$

differentiating

$\therefore y_2 = Ae^x + B\sin x + C\cos x$

$\therefore y_2' = Ae^x + B\cos x - C\sin x$

$\therefore y_2'' = Ae^x - B\sin x - C\cos x$

put y_2, y_2', y_2'' in (1)

$\therefore Ae^x - B\sin x - C\cos x - 2Ae^x - 2B\cos x + 2C\cos x - 3Ae^x - 3B\sin x - 3C\cos x = 2e^x - 10\sin x$

$\Rightarrow (A - 2A - 3A)e^x + (-B + 2C - 3B)\sin x + (-C - 2B - 3C)\cos x = 2e^x - 10\sin x$

$\Rightarrow -4Ae^x + (2C - 4B)\sin x + (-2B - 4C)\cos x = 2e^x - 10\sin x$

Comparing both sides:

$-4A = 2,\ 2C - 4B = -10,\ -2B - 4C = 0$

$\therefore A = -\frac{1}{2},\ B = 1,\ C = 2.$

$\therefore y_2 = -\frac{1}{2}e^x + \sin x + 2\cos x$

General sol^n is, $\boxed{y = y_1 + y_2}$

$\therefore y = c_1e^{-x} + c_2e^{3x} - \frac{1}{2}e^x + \sin x + 2\cos x$

comparing coefficients

$\therefore y = c_1e^{-x} + c_2e^{3x} - \frac{e^x}{2} + \sin x + 2\cos x \quad \rightarrow (3)$

$\therefore y' = -c_1e^{-x} + 3c_2e^{3x} - \frac{1}{2}e^x + \cos x - 2\sin x \quad \rightarrow (4)$

$\Rightarrow$ put $y(0) = 2$ in (3) & $y'(0) = 4$ in (4)

$② = c_1 e^0 + c_2 e^0 - \frac{e^0}{2} + \cancel{\sin} 0 + 2$ & $4 = -c_1 e^0 + 3c_2 e^0 - \frac{1}{2} e^0 + 1 - 0$

$\Rightarrow c_1 + c_2 + \frac{1}{2} = 0$ $\quad\Rightarrow 4 = -c_1 + 3c_2 - \frac{1}{2} + 1$

$\Rightarrow c_1 + c_2 = \frac{1}{2}$ $\quad\Rightarrow -c_1 + 3c_2 = \cancel{\frac{7}{2}}\ \frac{5}{2}$

$\therefore c_1 = -\frac{1}{4}$ $\quad c_2 = \frac{3}{4}$

$$\boxed{\therefore y = -\frac{1}{4} e^{-x} + \frac{3}{4} e^{3x} - \frac{1}{2} e^{x} + \sin x + 2\cos x}$$

(Ans)

Step ~~②~~ Step ③

+ Right Hand,

UC Function, UC set,

Step ①

(3) $y''-4y'+13y=8\sin 3x \longrightarrow (1)$

L.H.S:

$$y''-4y'+13y=0 \longrightarrow (2)$$

Let, $y=e^{mx}$

$$y'=me^{mx}$$

$$y''=m^2e^{mx}$$

$\Rightarrow m^2e^{mx}-4me^{mx}+13e^{mx}=0$

$\Rightarrow (m^2-4m+13)e^{mx}=0$

since $e^{mx}\neq 0$, eq^n will be:

$\Rightarrow m^2-4m+13=0$

$\Rightarrow m=\dfrac{-b\pm\sqrt{b^2-4ac}}{2a}$

$=\dfrac{4\pm\sqrt{16-52}}{2}$

$=\dfrac{4\pm\sqrt{-36}}{2} \Rightarrow \dfrac{4\pm\sqrt{36}\cdot\sqrt{-1}}{2}$

$=\cancel{2\pm\sqrt{-36}} \Rightarrow \dfrac{4\pm 6i}{2}$

$=\cancel{2\pm\sqrt{36}\cdot\sqrt{-1}} \Rightarrow \dfrac{\cancel{2}(2\pm 3i)}{\cancel{2}}$

$\Rightarrow \cancel{2\pm 6i\ [\text{complex}]} \Rightarrow 2\pm 3i$ [complex]

$\Rightarrow \boxed{y_1=e^{2x}(c_1\sin 3x+c_2\cos 3x)}$

R.H.S:

$S=\{\sin 3x,\cos 3x\}$

$y_2=A\sin 3x+B\cos 3x \longrightarrow (3)$

$\Rightarrow y_2'=3A\cos 3x-3B\sin 3x$

$\Rightarrow y_2''=-9A\sin x-9B\cos 3x$

$\Rightarrow -9A\sin 3x-9B\cos 3x$

$-12A\cos 3x+12B\sin 3x$

$+13A\sin 3x+13B\cos 3x=\cancel{0}\ 8\sin 3x$

$\Rightarrow (-9A+12B+13A)\sin 3x+(-9B-12A+13B)\cos 3x=\cancel{0}\ 8\sin 3x$

$\Rightarrow (4A+12B)\sin 3x+(4B-12A)\cos 3x=\cancel{0}\ 8\sin 3x$

comparing:

$\Rightarrow 4A+12B=8$	$4B-12A=0$
~~⇒ 4A +~~	$\Rightarrow 4B=12A$
$\Rightarrow A+3B=2$	$\Rightarrow B=3A$
$\Rightarrow A+3(3A)=2$	$\Rightarrow B=3/5$
$\Rightarrow 10A=2$	
$\Rightarrow A=\frac{1}{5}$	

$\Rightarrow \boxed{y_2=\frac{1}{5}\sin 3x+\frac{3}{5}\cos 3x}$

General eq^n:

$y=y_1+y_2$

$\Rightarrow \boxed{y=e^{2x}(c_1\sin 3x+c_2\cos 3x)+\frac{1}{5}\sin 3x+\frac{3}{5}\cos 3x}$

(Ans)

④ $y'' - 2y' - 8y = 4e^{2x} - 21e^{-3x} \longrightarrow ①$

L.H.S:

$y'' - 2y' - 8y = 0 \longrightarrow ②$

$\Rightarrow$ Let, $y = e^{mx}$

$y' = me^{mx}$

$y'' = m^2e^{mx}$

$\Rightarrow m^2e^{mx} - 2me^{mx} - 8e^{mx} = 0$

$\Rightarrow (m^2 - 2m - 8)e^{mx} = 0$

Since, $e^{mx} \neq 0$ eq^n will be

$\therefore m^2 - 2m - 8 = 0$

$\Rightarrow m^2 - 4m + 2m - 8 = 0$

$\Rightarrow m(m-4) + 2(m-4) = 0$

$\Rightarrow (m+2)(m-4) = 0$

$\Rightarrow m = -2, 4$ [distinct]

$\Rightarrow \boxed{Y_1 = C_1 m^{-2x} + C_2 m^{4x}}$

R.H.S:

$S_1 = \{e^{2x}\}, S_2 = \{e^{-3x}\}$

UC set,

$S = \{e^{2x}, e^{-3x}\}$

$Y_2 = Ae^{2x} + Be^{-3x} \longrightarrow ③$

$Y_2' = 2Ae^{2x} - 3Be^{-3x}$

$Y_2'' = 4Ae^{2x} + 9Be^{-3x}$

$\Rightarrow 4Ae^{2x} + 9Be^{-3x}$

$-4Ae^{2x} + 6Be^{-3x}$

$-8Ae^{2x} - 8Be^{-3x} = 4e^{x} - 21e^{-3x}$

$\Rightarrow (4A - 4A - 8A)e^{2x} + (9B + 6B - 8B)e^{-3x} =$

$\Rightarrow -8Ae^{2x} + 7Be^{-3x} = 4e^{2x} - 21e^{-3x}$

Comparing:

$-8A = 4$	$7B = -21$
$\Rightarrow A = -\frac{1}{2}$	$B = -3$

$\Rightarrow \boxed{Y_2 = -\frac{1}{2}e^{2x} - 3e^{-3x}}$

General eqn:

$Y = Y_1 + Y_2$

$\Rightarrow \boxed{Y = C_1 m^{-2x} + C_2 m^{4x} - \frac{1}{2}e^{2x} - 3e^{-3x}}$

(Ans)

(5) $y'' - 4y' + 13y = 18e^{-2x}$ → (1)

L.H.S:

$y'' - 4y' + 13y = 0$ → (2)

Let, $y = e^{mx}$

$y' = me^{mx}$

$y'' = m^2 e^{mx}$

$\therefore m^2 e^{mx} - 4me^{mx} + 13e^{mx} = 0$

$\Rightarrow (m^2 - 4m + 13) e^{mx} = 0$

Since $e^{mx} \neq 0$, eq^n will be:

$\therefore m^2 - 4m + 13 = 0$

$\Rightarrow m = \frac{4 \pm \sqrt{-36}}{2}$

$m = 2 \pm 3i$ [complex]

$\boxed{y_1 = e^{2x}(\cancel{c_1 e^{6x}} + c_1 \sin 3x + c_2 \cos 3x)}$

R.H.S:

$S = \{e^{-2x}\}$

$\Rightarrow y_2 = Ae^{-2x}$

$\Rightarrow y_2' = -2Ae^{-2x}$

$\Rightarrow y_2'' = 4Ae^{-2x}$

$\Rightarrow 4Ae^{-2x} + 8Ae^{-2x} + 13Ae^{-2x} = 18e^{-2x}$

$\Rightarrow 25Ae^{-2x} = 18e^{-2x}$

Comparing

$25A = 18$

$A = 18/25$

$\Rightarrow \boxed{y_2 = \frac{18}{25} e^{-2x}}$

→ General

$y = y_1 + y_2$

$\Rightarrow \boxed{y = e^{2x}\{c_1 \sin(3x) + c_2 \cos(3x)\} + \frac{18}{25} e^{-2x}}$

(A)

2) $y'' - 4y' + 3y = 9x^2 + 4$, $y(0) = 6$, $y'(0) = 3$ $\rightarrow (1)$

L.H.S:

$y'' - 4y' + 3y = 0 \rightarrow (2)$

Let, $y = e^{mx}$

$y' = me^{mx}$

$y'' = m^2 e^{mx}$

$\Rightarrow m^2 e^{mx} - 4me^{mx} + 3e^{mx} = 0$

$\Rightarrow (m^2 - 4m + 3) e^{mx} = 0$

$\because$ Since $e^{mx} \neq 0$, auxillary eqn will be

$\Rightarrow m^2 - 4m + 3 = 0$

$\Rightarrow m^2 - 3m - m + 3 = 0$

$\Rightarrow m(m-3) - 1(m-3) = 0$

$\Rightarrow (m-3)(m-1) = 0$

$\Rightarrow m = 3, 1$ [distinct]

$\boxed{y_1 = c_1 e^{3x} + c_2 e^{x}}$

R.H.S:

$S = \{x^2, x, 1\}$

$y_2 = Ax^2 + Bx + C \rightarrow (3)$

$\Rightarrow y_2' = 2Ax + B$

$\Rightarrow y_2'' = 2A$

$\Rightarrow 2A - 8Ax - 4B + 3Ax^2 + 3Bx + 3C = 9x^2 + 4$

$\Rightarrow 3Ax^2 + (-8A + 3B)x + (2A - 4B + 3C) = 9x^2 + 4$

Comparing,

$3A = 9$	$-8A + 3B = 0$	$2A - 4B + 3C = 4$
$A = 3$	$\Rightarrow -8\times3 + 3B = 0$	$\Rightarrow 6 - 32 + 3C = 0$
	$\Rightarrow 3B = 24$	$\Rightarrow 3C = 26 + 4$
	$\Rightarrow B = 8$	$\Rightarrow C = 10$

$\boxed{y_2 = 3x^2 + 8x + 10}$

General eqn,

$y = y_1 + y_2$

$\Rightarrow y = c_1 e^{3x} + c_2 e^{x} + 3x^2 + 8x + 10$

$\Rightarrow y' = 3c_1 e^{3x} + c_2 e^{x} + 6x + 8$

$\Rightarrow 3 = 3c_1 e^0 + c_2 e^0 + 0 + 8$

$\Rightarrow 3c_1 + c_2 = -5$

$6 = c_1 + c_2 + 10$

$\Rightarrow c_1 + c_2 = -4$

$\boxed{c_1 = -\frac{1}{2},\ c_2 = -\frac{7}{2}}$

$\Rightarrow \boxed{y = -\frac{1}{2}e^{3x} - \frac{7}{2}e^{x} + 3x^2 + 8x + 10}$

(Ans)